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Semiconductor memory device and manufacturing method of semiconductor memory device

Abstract

There are provided a semiconductor memory device and a manufacturing method of a semiconductor memory device. The manufacturing method of the semiconductor memory device includes: stacking a plurality of first material layers and a plurality of second material layers over a preliminary doped semiconductor structure; forming a blocking insulating layer, a data storage layer, a tunnel insulating layer, and a channel layer, which penetrate the plurality of first and second material layers, and extend to the inside of the preliminary doped semiconductor structure; forming a slit penetrating the plurality of first and second material layers; forming a protective structure as a double layer or a single layer on a sidewall of the slit; and forming a doped channel contact layer which penetrates a portion of the preliminary doped semiconductor structure in a direction intersecting the channel layer, and is in contact with the channel layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority under 35 U.S.C. § 119(a) to Korean patent application number 10-2022-0085345 filed on Jul. 12, 2022, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated by reference herein.

BACKGROUND

1. Technical Field

(2) The present disclosure generally relates to a semiconductor memory device and a manufacturing method of a semiconductor memory device, and more particularly, to a three-dimensional semiconductor memory device and a manufacturing method of a three-dimensional semiconductor memory device.

2. Related Art

(3) A semiconductor memory device may include a plurality of memory cells capable of storing

data. A plurality of memory cells of a three-dimensional semiconductor memory device may be three-dimensionally arranged. In the three-dimensional semiconductor memory device, the plurality of memory cells may be connected in series by a channel structure penetrating a gate stack structure. The channel layer of the three-dimensional semiconductor memory device may be electrically connected to a doped semiconductor structure overlapping with the gate stack structure.

SUMMARY

(4) In accordance with an embodiment of the present disclosure, there may be provided a method of manufacturing a semiconductor memory device, the method may include: alternately stacking a plurality of first material layers and a plurality of second material layers in a first direction over a preliminary doped semiconductor structure; forming a hollowed blocking insulating layer penetrating the plurality of first material layers and the plurality of second material layers, the hollowed blocking insulating layer extending to the inside of the preliminary doped semiconductor structure; forming a plurality of data storage patterns on levels where the plurality of second material layers are disposed, wherein an inner wall of the hollowed blocking insulating layer is partially covered by the plurality of data storage patterns; forming a hollowed tunnel insulating layer to cover the plurality of data storage patterns and the inner wall of the hollowed blocking insulating layer; forming a channel layer inside the hollowed tunnel insulating layer; forming a slit penetrating the plurality of first material layers and the plurality of second material layers; forming a protective structure as one of a double layer and a single layer on a sidewall of the slit; and forming a doped channel contact layer penetrating a portion of the preliminary doped semiconductor structure in a direction intersecting the channel layer, the doped channel contact layer being in contact with the channel layer.

(5) In accordance with an embodiment of the present disclosure, there may be provided a semiconductor memory device including: a doped semiconductor structure; a gate stack structure including a plurality of conductive layers stacked in a first direction over the doped semiconductor structure; a channel layer including a first part and a second part extending in a first direction from the first part to penetrate the gate stack structure; a first tunnel insulating pattern extending along an outer wall of the second part of the channel layer; a plurality of data storage patterns interposed between the plurality of conductive layers and the first tunnel insulating pattern; and a first blocking insulating pattern interposed between each of the plurality of data storage pattern and the gate stack structure, wherein at least a portion of the first part is located inside the doped semiconductor structure, wherein the doped semiconductor structure includes a contact surface in contact with the channel layer, and wherein the first blocking insulating pattern extends to a first region between a lowermost conductive layer adjacent to the doped semiconductor structure among the plurality of conductive layer and the contact surface, and is in contact with the first tunnel insulating pattern in the first region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various examples of embodiments will now be described more fully hereinafter with reference to the accompanying drawings; however, they may be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be enabling to those skilled in the art.

(2) In the drawing figures, dimensions may be exaggerated for clarity of illustration. It will be understood that when an element is referred to as being “between” two elements, it may be the only element between the two elements, or additional intervening elements may also be present. Like reference numerals refer to like elements throughout the drawings.

(3) FIG. 1 is a diagram schematically illustrating a semiconductor memory device in accordance

with an embodiment of the present disclosure.

(4) FIG. 2 is a circuit diagram of a memory cell array in accordance with an embodiment of the present disclosure.

(5) FIG. 3 is a perspective view illustrating a semiconductor memory device in accordance with an embodiment of the present disclosure.

(6) FIG. 4 is an enlarged sectional view of a portion of the semiconductor memory device shown in FIG. 3.

(7) FIGS. 5A, 5B, 5C, and 5D are sectional views illustrating some processes of a manufacturing method of a semiconductor memory device in accordance with an embodiment of the present disclosure.

(8) FIGS. 6A, 7A, 8A, and 9A are enlarged sectional views illustrating an embodiment of subsequent processes performed after the process shown in FIG. 5D.

(9) FIGS. 6B, 7B, 8B, and 9B are enlarged sectional views illustrating an embodiment of the subsequent processes performed after the process shown in FIG. 5D.

(10) FIGS. 10A and 10B are sectional views illustrating an embodiment of subsequent processes performed after the process shown in FIG. 9A or 9B.

(11) FIG. 11 is a block diagram illustrating a configuration of a memory system in accordance with an embodiment of the present disclosure.

(12) FIG. 12 is a block diagram illustrating a configuration of a computing system in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

(13) Specific structural and functional descriptions disclosed herein are merely illustrative for the purpose of describing embodiments according to the concept of the present disclosure.

Embodiments according to the concept of the present disclosure can be implemented in various forms, and they should not be construed as being limited to the specific embodiments set forth herein.

(14) It will be understood that, although the terms “first”, “second”, etc. may be used herein to describe various elements, these elements are not limited by these terms. These terms are used for distinguishing one element from another element and not to suggest a number or order of elements. It will be understood that when a structure, element, or layer etc., is referred to as being “on,” “connected to” or “coupled to” another structure, element, or layer etc., it can be directly on, connected or coupled to the other structure, element, or layer etc., or intervening structures, elements, or layers etc., may be present. In contrast, when a structure, element, or layer etc., is referred to as being “directly on,” “directly connected to” or “directly coupled to” another structure, element, or layer etc., there are no intervening structures, elements, or layers etc., present. In some embodiments, it will be understood that when a layer, structure, or element etc., is referred to as penetrating another layer, structure, or element etc., it can partially penetrate the other layer, structure, or element etc., by extending into a portion of the other layer, structure, or element etc., or completely penetrate the other layer, structure, or element etc., by passing through the entire layer, structure, or element etc.

(15) Various embodiments provide a semiconductor memory device and a manufacturing method of a semiconductor memory device, which may simplify a manufacturing process.

(16) FIG. 1 is a diagram schematically illustrating a semiconductor memory device in accordance with an embodiment of the present disclosure.

(17) Referring to FIG. 1, the semiconductor memory device may include a doped semiconductor structure DPS, a plurality of memory cell arrays MCA1 and MCA2, and a plurality of bit lines BL. The plurality of memory cell arrays MCA1 and MCA2 may be disposed between the doped semiconductor structure DPS and the plurality of bit lines BL.

(18) Each of the plurality of memory cell arrays MCA1 and MCA2 may include a plurality of memory cell strings connected to the doped semiconductor structure DPS. Each memory cell string

may be connected to a plurality of conductive layers of a gate stack structure corresponding thereto. The plurality of memory cell arrays MCA1 and MCA2 may include a first memory cell array MCA1 and a second memory cell array MCA2 adjacent to each other on a plane parallel to a top surface of the doped semiconductor structure DPS. A gate stack structure of the first memory cell array MCA1 and a gate stack structure of the second memory cell array MCA2 may be isolated from each other by a slit.

(19) Each bit line BL may be connected to a memory cell string of the first memory cell array MCA1 and a memory cell string of the second memory cell array MCA2. Each memory cell string may include a channel layer connected to a bit line BL corresponding thereto and the doped semiconductor structure DPS.

(20) Although not shown in the drawing, the semiconductor memory device may further include a peripheral circuit structure for controlling the plurality of memory cell arrays MCA1 and MCA2, the plurality of bit lines BL, and the doped semiconductor structure DPS. In an embodiment, the peripheral circuit structure may be disposed adjacent to the plurality of bit lines BL. An interconnection may be disposed between the peripheral circuit structure and the plurality of bit lines BL, or an interconnection and a bonding structure may be disposed between the peripheral circuit structure and the plurality of bit lines BL. In another embodiment, the peripheral circuit structure may be disposed adjacent to the doped semiconductor structure DPS. An interconnection may be disposed between the peripheral circuit structure and the doped semiconductor structure DPS, or an interconnection and a bonding structure may be disposed between the peripheral circuit structure and the doped semiconductor structure DPS.

(21) A manufacturing process of the semiconductor memory device may be various. In an embodiment, a process of forming the doped semiconductor structure DPS, a process of forming the plurality of memory cell arrays MCA1 and MCA2, and a process of forming the plurality of bit lines BL may be performed on the peripheral circuit structure (not shown). In another embodiment, a first structure including the plurality of memory cell arrays MCA1 and MCA2 may be formed separately from the second structure including the peripheral circuit structure (not shown). The first structure and the second structure may be structurally connected to each other through a bonding structure. The bonding structure may include a first bonding insulating layer disposed at the first structure, a second bonding insulating layer disposed at the second structure, a first conductive bonding pad inside the first bonding insulating layer, and a second conductive bonding pad inside the second bonding insulating layer. The first bonding insulating layer may be coupled to the second bonding insulating layer, and the first conductive bonding pad may be coupled to the second conductive bonding pad. The first conductive bonding pad may be electrically connected to the peripheral circuit structure via an interconnection corresponding thereto, and the second conductive bonding pad may be electrically connected to a memory cell array corresponding thereto via an interconnection corresponding thereto.

(22) FIG. 2 is a circuit diagram of a memory cell array in accordance with an embodiment of the present disclosure.

(23) Referring to FIG. 2, a plurality of memory cell strings CS of the memory cell array may be connected in parallel to a common source line CSL. The common source line CSL may be connected to the plurality of memory cell strings CS via the doped semiconductor structure DPS shown in FIG. 1.

(24) Each memory cell string CS may include at least one source select transistor SST, a plurality of memory cells MC1 to MCn, and at least one drain select transistor DST. The plurality of memory cells MC1 to MCn may be connected in series between the source select transistor SST and the drain select transistor DST. The source select transistor SST, the plurality of memory cells MC1 to MCn, and the drain select transistor DST may be connected in series by a channel layer.

(25) Each memory cell string CS may be connected to a bit line corresponding thereto among a plurality of bit lines BL. The common source line CSL and the plurality of bit lines BL may be

connected to channel layers of the plurality of memory cell strings CS.

(26) The plurality of memory cells MC1 to MCn of the memory cell string CS may be connected to the common source line CSL via the source select transistor SST and the doped semiconductor structure DPS shown in FIG. 1. The plurality of memory cells MC1 to MCn of the memory cell string CS may be connected to a bit line BL corresponding thereto via the drain select transistor DST.

(27) The memory cell string CS may be connected to a source select line SSL, a plurality of word lines WL1 to WLn, and a drain select line DSL1 or DSL2. The source select line SSL may be used as a gate electrode of the source select transistor SST. The plurality of word lines WL1 to WLn may be used as gate electrodes of the plurality of memory cells MC1 to MCn. The drain select line DSL1 or DSL2 may be used as a gate electrode of the drain select transistor DST.

(28) The plurality of memory cell strings CS may be controlled by each of the plurality of word lines WL1 to WLn. Two or more memory cell strings CS may be controlled by each bit line BL. In an embodiment, one memory cell string CS of a first memory cell string group CS[A] and one memory cell string CS of a second memory cell string group CS[B] may be connected to each bit line BL. The first memory cell string group CS[A] and the second memory cell string group CS[B] may be individually controlled by drain select lines isolated from each other or source select lines isolated from each other. In an embodiment, the first memory cell string group CS[A] may be connected to a first drain select line DSL1, and the second memory cell string group CS[B] may be connected to a second drain select line DSL2. The first memory cell string group CS[A] and the second memory cell string group CS[B] may be connected to the same source select line SSL. However; the embodiments of the present disclosure is not limited thereto. In other embodiments, two or more memory cell string groups connected to the same bit line BL may be connected to the same drain select line, and be individually connected to two or more source select lines isolated from each other. In still other embodiments, two or more memory cell string groups connected to the same bit line BL may be individually connected to two or more drain select lines isolated from each other, and be individually connected to two or more source select lines isolated from each other.

(29) An operating voltage for precharging a channel layer of a memory cell string CS corresponding to each bit line BL may be applied to the bit line BL. The bit line BL may be connected to the channel layer of the memory cell string CS through a bit line connection structure.

(30) An operating voltage for discharging a potential of the channel layer of the memory cell string CS may be applied to the common source line CSL.

(31) FIG. 3 is a perspective view illustrating a semiconductor memory device in accordance with an embodiment of the present disclosure. FIG. 4 is an enlarged sectional view of a portion of the semiconductor memory device shown in FIG. 3.

(32) Referring to FIGS. 3 and 4, the semiconductor memory device may include a plurality of gate stack structures GST disposed over a doped semiconductor structure 100 and a plurality of channel layers 131 which penetrate the plurality of gate stack structures GST and extend to the inside of the doped semiconductor structure 100. A first tunnel insulating pattern 125A, a first blocking insulating pattern 121A, and a plurality of data storage patterns 123P may be interposed between each gate stack structure GST and a channel layer corresponding thereto. A plurality of bit lines 187 may be disposed above the plurality of gate stack structures GST.

(33) The doped semiconductor structure 100 may include a contact surface CTS in contact with the channel layer 131. More specifically, the doped semiconductor structure 100 may include a lower layer 101, a doped channel contact layer 161, and an upper layer 109. The doped channel contact layer 161 may include a sidewall 161SW corresponding to the contact surface CTS. The doped channel contact layer 161 may be disposed on the lower layer 101. The upper layer 109 may be disposed between the gate stack structure GST and the doped channel contact layer 161. The upper layer 109 may be spaced apart from the lower layer 101 in a first direction DR1 with the doped

channel contact layer **161** interposed therebetween. The upper layer **109** may be omitted in some embodiments.

(34) Each of the lower layer **10** the doped channel contact layer **161**, and the upper layer **109** may include a doped semiconductor layer. Each of the lower layer **101**, the doped channel contact layer **161**, and the upper layer **109** may include at least one of an n-type impurity and a p-type impurity.

(35) The semiconductor memory device may include a source contact structure **163** inside a slit **153**. The source contact structure **163** may be in contact with the doped channel contact layer **161** and may extend in the first direction **DR1**. The source contact structure **163** may be formed of the same material as the doped channel contact layer **161**, or be formed of various conductive materials including a metal. The source contact structure **163** may be provided to electrically connect the doped semiconductor structure **100** to the common source line CSL shown in FIG. 2.

(36) The plurality of gate stack structures GST may be disposed between the doped semiconductor structure **100** and the plurality of bit lines **187**. The plurality of gate stack structures GST may be partitioned by the slit **153**. FIGS. 3 and 4 illustrate a portion of each gate stack structure GST adjacent to slit **153**. Each of the plurality of gate stack structures GST may include a plurality of conductive layers **155**. The plurality of conductive layers **155** may be provided as a plurality of gate electrodes of a memory cell array corresponding thereto. In an embodiment, the semiconductor memory device may include a first memory cell array **10A** and a second memory cell array **10B**, which are disposed at both sides of the slit **153**. A gate stack structure GST of each of the first memory cell array **10A** and the second memory cell array **10B** may include a plurality of conductive layers **155** stacked over the doped semiconductor structure **100**. In an embodiment, a gate insulating layer GI may be located between the plurality of conductive layers **155** and the doped semiconductor structure **100** as, for example, shown in FIG. 4. A sidewall insulating layer **157** may be disposed between the source contact structure **163** and the gate stack structure GST. The source contact structure **163** may be insulated from the plurality of conductive layers **155** of the gate stack structure GST by the sidewall insulating layer **157**.

(37) The plurality of conductive layers **155** may be stacked to be spaced apart from each other in the first direction **DR1**. Each conductive layer **155** may have a flat plate shape extending in a second direction **DR2** and a third direction **DR3**. In an embodiment, the second direction **DR2** and the third direction **DR3** may respectively correspond to an X-axis direction and a Y-axis direction, and the first direction **DR1** may correspond to a Z-axis direction.

(38) The plurality of conductive layers **155** may be used as the source select line SSL, the plurality of word lines WL1 to WLn, and the drain select line DSL1 or DSL2, which are shown in FIG. 2. At least one conductive layer **155** adjacent to the doped semiconductor structure **100** among the plurality of conductive layers **155** may be used as the source select line. In an embodiment, a lowermost conductive layer **155B** adjacent to the doped semiconductor structure **100** among the plurality of conductive layers **155** may be used as the source select line. At least one conductive layer adjacent to the bit line **187** among the plurality of conductive layers **155** may be used as the drain select line. In an embodiment, an uppermost conductive layer **155T** adjacent to the bit line **187** among the plurality of conductive layers **155** may be used as the drain select line. The other conductive layers between the conductive layer (e.g., **155T**) used as the drain select line and the conductive layer (e.g., **155B**) used as the source select line may be used as the word lines. The lowermost conductive layer **155B** may be spaced apart from the doped semiconductor structure **100** by the gate insulating layer GI.

(39) The gate stack structure GST may further include a plurality of interlayer insulating layers **111** alternately disposed in the first direction **DR1** with the plurality of conductive layers **155**. A plurality of conductive layers **155** adjacent to each other in the first direction **DR1** may be insulated from each other by a plurality of interlayer insulating layers **111**. Each conductive layer **155** may include at least one of a doped semiconductor layer, a metal layer, and a conductive metal nitride layer. The doped semiconductor layer may include a doped silicon layer. The metal layer may

include tungsten, copper, molybdenum, and the like. The conductive metal nitride layer may include titanium nitride, tantalum nitride, and the like.

(40) The channel layer **131** may be formed of a semiconductor material such as silicon or germanium. In an embodiment, the channel layer **131** may include a substantially intrinsic semiconductor. A partial region of the channel layer **131**, which forms the contact surface CTS with the doped semiconductor structure **100**, and a partial region of the channel layer **131**, which is adjacent thereto, may be defined as a source junction. The source junction may include, as a majority carrier, an impurity having the same conductivity type as the channel contact layer **161**. In an embodiment, the source junction may include an n-type impurity. In addition, another partial region of the channel layer **131** may be defined as a drain junction. A top end of the channel layer **131**, which is adjacent to the bit line **187**, may be defined as the drain junction. The drain junction may include at least one of an n-type impurity and a p-type impurity. In an embodiment, the drain junction may include the n-type impurity as a majority carrier.

(41) The channel layer **131** may be formed in various structures such as a pillar structure and a hollow structure. When the channel layer **131** is formed in a hollow structure, a core insulating layer **133** and a capping semiconductor layer **135** may be disposed inside the hollowed portion of the channel layer **131**, as for example, shown in FIG. 3. The capping semiconductor layer **135** may be disposed at the top end of the hollowed portion of the channel layer **131** adjacent to the bit line **187**, and overlap with the core insulating layer **133**. The capping semiconductor layer **135** may include an impurity having the same conductivity type as the drain junction. In an embodiment, the capping semiconductor layer **135** may include an n-type impurity.

(42) The channel layer **131** may be used as a channel region of a memory cell string corresponding thereto, and a potential of channels inside the channel layer **131** may be changed by a voltage applied to the plurality of conductive layers **155**.

(43) The channel layer **131** may include a first part **P1** inside the doped semiconductor structure **100** and a second part **P2** inside the gate stack structure GST. The first part **P1** may penetrate the upper layer **109** and the doped channel contact layer **161** of the doped semiconductor structure **100**, and extend to the inside of the lower layer **101** of the doped semiconductor structure **100**. Also, the first part **P1** may penetrate the gate insulating layer GI. The first part **P1** may form the contact surface CTS with the doped channel contact layer **161**. Accordingly, the doped semiconductor structure **100** may be electrically connected to the channel layer **131** through the channel contact layer **161**. The second part **P2** may extend in the first direction DR1 from the first part **P1** to penetrate the gate stack structure GST.

(44) The first tunnel insulating pattern **125A** may be interposed between the channel layer **131** and the gate stack structure GST, to extend along an outer wall of the second part **P2** of the channel layer **131**. The first tunnel insulating pattern **125** may be formed of an insulating material through which charges can tunnel. In an embodiment, the first tunnel insulating pattern **125A** may include silicon oxide.

(45) The plurality of data storage patterns **123P** may be interposed between the plurality of conductive layers **155** and the first tunnel insulating pattern **125A**. The plurality of data storage patterns **123P** may be formed of a material layer capable of storing data changed using Fowler-Nordheim tunneling. To this end, each data storage pattern **123P** may include at least one of a charge trap layer, a floating gate layer, and an insulating layer including a conductive nano dot. The charge trap layer may include silicon nitride, and the floating gate layer may include a silicon layer.

(46) The plurality of data storage patterns **123P** may be divided into a pattern used as a charge storage layer of a memory cell and a pattern as a gate insulating layer. A pattern corresponding to the conductive layer used as the word line among the plurality of data storage patterns **123P** may be used as the charge storage layer, and a pattern corresponding to the conductive layers used as the drain select line and the source select line among the plurality of data storage patterns **123P** may be used as the gate insulating layer.

(47) The plurality of data storage patterns **123P** may be disposed between the doped semiconductor structure **100** and the first part **P1** of the channel layer **131**. In an embodiment, each of the plurality of data storage patterns **123P** may be respectively disposed at levels at which the plurality of conductive layers **155** of the gate stack structure **GST** are disposed. To this end, the plurality of interlayer insulating layers **111** may further protrude toward the channel layer **131** than the plurality of conductive layers **155**. The plurality of data storage patterns **123P** may be disposed between protrusion parts of the plurality of interlayer insulating layers **111**, and be isolated from each other in the first direction **DR1** by the plurality of interlayer insulating layers **111**. The structure for disposing the plurality of data storage patterns **123P** between the doped semiconductor structure **100** and the channel layer **131** is not limited to the above-described embodiments. Although not shown in the drawings, the plurality of data storage patterns **123P** formed of an insulating material may extend between each of the plurality of interlayer insulating layers **111** and the gate insulating layer **GI**, and the first tunnel insulating pattern **125A** to be connected to each other over the doped semiconductor structure **100**. Hereinafter, for convenience of description, the present disclosure is described based on various embodiments in which the plurality of data storage patterns **123P** are spaced apart from each other in the first direction **DR1**.

(48) The first blocking insulating pattern **121A** may be interposed between each of the plurality of data storage patterns **123P** and the gate stack structure **GST**. The first blocking insulating pattern **121A** may extend to a first region **AR1** between the contact surface **CTS** and the lowermost conductive layer **155B**. The first blocking insulating pattern **121A** may be in contact with the first tunnel insulating pattern **125** in the first region **AR1**. More specifically, the first blocking insulating pattern **121A** and the first tunnel insulating pattern **125A** may extend between the first part **P1** of the channel layer **131** and the upper layer **109** of the doped semiconductor structure **100**, and be in contact with each other between the first part **P1** of the channel layer **131** and the upper layer **109**. The first blocking insulating pattern **121A** may extend to a second region **AR2** between conductive layers **155** adjacent to each other in the first direction **DR1**. When the plurality of data storage patterns **123P** are disposed to be spaced apart from each other in the first direction **DR1**, the first blocking insulating pattern **121A** and the first tunnel insulating pattern **125A** may be in contact with each other in the second region **AR2**. That is, the first blocking insulating pattern **121A** and the first tunnel insulating pattern **125A** may be in contact with each other between the plurality of interlayer insulating layers **111** and the second part **P2** of the channel layer **131**.

(49) The first blocking insulating pattern **121A** may include an insulating material capable of blocking charges. In an embodiment, the first blocking insulating pattern **121A** may include at least one of silicon oxide and metal oxide. The first blocking insulating pattern **121A** may extend to the first region **AR1** between the contact surface **CTS** and the lowermost conductive layer **155B**.

(50) The semiconductor memory device may further include a second blocking insulating pattern **121B** and a second tunnel insulating pattern **125B**, which are interposed between the lower layer **101** of the doped semiconductor structure **100** and the first part **P1** of the channel layer **131**. The second blocking insulating pattern **121B** may be formed of the same material as the first blocking insulating pattern **121A** and may be spaced apart from the first blocking insulating pattern **121A** by the doped channel contact layer **161**. The second tunnel insulating pattern **125B** may be formed of the same material as the first tunnel insulating pattern **125A** and may be spaced apart from the first tunnel insulating pattern **125A** by the doped channel contact layer **161**.

(51) The second tunnel insulating pattern **125B** may be interposed between the first part **P1** of the channel layer **131** and the lower layer **101**. The second blocking insulating pattern **121B** may be interposed between the second tunnel insulating pattern **125B** and the lower layer **101**. The second tunnel insulating pattern **125B** and the second blocking insulating pattern **121B** may be in contact with each other.

(52) The plurality of bit lines **187** may be conductive patterns. The plurality of bit lines **187** may extend in a direction intersecting the plurality of gate stack structures **GST**. In an embodiment, the

plurality of gate stack structures GST may be spaced apart from each other in the second direction DR2 by the slit **153** extending in the third direction DR3, and the plurality of bit lines **187** may extend in the second direction DR2. The plurality of bit lines **187** may be spaced apart from the source contact structure **163**. Although not shown in the drawings, each of the plurality of bit lines **187** may be connected to a channel layer **131** and a capping semiconductor layer **135**, which correspond thereto through a bit line connection structure formed of a conductive material.

(53) A protective structure may be formed on a sidewall of the slit **153** in a process of manufacturing the above-described semiconductor memory device. The protective structure may be designed by considering material layers interposed between the channel layer **131** and the doped semiconductor structure **100**. In an embodiment of the present disclosure, oxide-based material layers such as the first tunnel insulating pattern **125A**, the first blocking insulating pattern **121A**, the second tunnel insulating pattern **125B**, and the second blocking insulating pattern **121** may be interposed between the doped semiconductor structure **100** and the channel layer **131**, and a material having an etch selectivity with respect to the data storage pattern **123P** may be excluded in the protective structure. In consideration of this, although the protective structure is simplified as a double layer or a single layer, various embodiments of the present disclosure may provide a manufacturing method of the semiconductor memory device, which can stably perform a process. A manufacturing method of a semiconductor memory device in accordance with an embodiment of the present disclosure will be described in more detail with reference to the following drawings.

(54) FIGS. 5A to 5D are sectional views illustrating some processes of a manufacturing method of a semiconductor memory device in accordance with an embodiment of the present disclosure.

(55) Referring to FIG. 5A, a preliminary doped semiconductor structure **300** may be formed on a lower structure (not shown). Although not shown in the drawing, the lower structure may be a peripheral circuit structure or a sacrificial substrate.

(56) The preliminary doped semiconductor structure **300** may include a lower layer **301**, a sacrificial layer **305**, and an upper layer **309**, which are stacked in the first direction DR1. The preliminary doped semiconductor structure **300** may further include a lower protective layer **303** between the lower layer **301** and the sacrificial layer **305** and an upper protective layer **307** between the sacrificial layer **305** and the upper layer **309**. The lower protective layer **303** and the upper protective layer **307** may be formed of an oxide-based material. The sacrificial layer **305** may be formed of a material having an etch selectivity with respect to the lower protective layer **303** and the upper protective layer **307**. In an embodiment, the sacrificial layer **305** may be formed of an undoped silicon layer. When the sacrificial layer **305** has an etch selectivity with respect to the lower protective layer **303** and the upper protective layer **307**, the lower protective layer **303** and the upper protective layer **307** may be omitted. In an embodiment, the sacrificial layer **305** may include a metal having an etch selectivity with respect to the lower protective layer **303** and the upper protective layer **307**.

(57) The lower layer **301** may be formed of a doped semiconductor layer. The upper layer **309** may be formed of a material having an etch selectivity with respect to a preliminary gate stack structure **310** to be formed subsequently. In an embodiment, the upper layer **309** may be formed of a semiconductor layer such as silicon.

(58) Subsequently, a plurality of first material layers **201** and **311** and a plurality of second material layers **313** may be alternately stacked in the first direction DR1 over the preliminary doped semiconductor structure **300**. The plurality of first material layers **201** and **311** may include a gate insulating layer **201** and a plurality of interlayer insulating layers **311**. The gate insulating layer **201** and the plurality of interlayer insulating layers **311** may include oxide such as a silicon oxide layer. The gate insulating layer **201** may be formed to have a thickness thinner than a thickness of each of the plurality of interlayer insulating layers **311**. The plurality of second material layers **313** may be formed of a material different from the material of the plurality of first material layers **201** and **311**. In an embodiment, the second material layer **313** may be a conductive layer including at least one

of a doped semiconductor layer, a metal layer, and a conductive metal nitride layer. In another embodiment, the second material layer **313** may be provided as a sacrificial material having an etch selectivity with respect to the plurality of first material layers **201** and **311**. The second material layer **313** may be formed of silicon nitride. The plurality of interlayer insulating layers **311** and the plurality of second material layers **313** may be provided as the preliminary gate stack structure **310**. (59) Referring to FIG. 5B, a plurality of holes H may be formed. The plurality of holes H may penetrate the plurality of first material layers **201** and **311** and the plurality of second material layers **313** and may extend to the inside of the preliminary doped semiconductor structure **300**. To this end, each of the plurality of first material layers **201** and **311**, the plurality of second material layers **313**, the upper layer **309**, the upper protective layer **307**, the sacrificial layer **305**, and the lower protective layer **303** may be locally etched through an etching process using a photolithography process. The etching process may be controlled such that the plurality of holes H may extend to the inside of the lower layer **301**.

(60) Subsequently, a plurality of recess regions R may be formed between the plurality of first material layers **201** and **311** by recessing the plurality of second material layers **313** exposed through the plurality of holes H, using a selective etching process.

(61) Referring to FIG. 5C, a hollowed blocking insulating layer **321** may be formed along surfaces of each hole H and a plurality of recess regions (R shown in FIG. 5B). The hollowed blocking insulating layer **321** may be conformally formed such that the plurality of recess regions R shown in FIG. 5B may be opened. Subsequently, a hollowed data storage layer **323** may be formed inside the hollowed blocking insulating layer **321** such that the plurality of recess regions (R shown in FIG. 5B) opened by the hollowed blocking insulating layer **321** are filled.

(62) The hollowed blocking insulating layer **321** may include oxide. In an embodiment, the hollowed blocking insulating layer **321** may include at least one of silicon oxide and metal oxide. The hollowed data storage layer **323** may include at least one of insulating layers including a charge trap layer, a floating gate layer, and a conductive nano dot. The charge trap layer may include silicon nitride, and the floating gate layer may include a silicon layer.

(63) Referring to FIG. 5D, the hollowed data storage layer **323** shown in FIG. 5C may be isolated into a plurality of data storage patterns **323P** by removing a portion of the hollowed data storage layer **323** shown in FIG. 5C inside the hole H. The plurality of data storage patterns **323P** may respectively correspond to the second material layers **313**, and remain inside the plurality of recess regions R shown in FIG. 5B.

(64) Subsequently, a hollowed tunnel insulating layer **325** may be formed inside the hollowed data storage layer isolated into the plurality of data storage patterns **323P**. The hollowed tunnel insulating layer **325** may include oxide having a dielectric constant lower than a dielectric constant of the hollowed blocking insulating layer **321**. In an embodiment, the hollowed tunnel insulating layer **325** may include silicon oxide.

(65) Subsequently, a channel layer **331** may be formed inside the hollowed tunnel insulating layer **325**. The channel layer **331** may include a semiconductor layer such as silicon or germanium. When a central region of the hollowed tunnel insulating layer **325** is opened by the channel layer **331**, a core insulating layer **333** and a capping semiconductor layer **335** may be formed in the central region of the hollowed tunnel insulating layer **325**.

(66) According to the above-described process, the hollowed blocking insulating layer **321** may include a first part **321P1** inside the preliminary doped semiconductor structure **300** and a second part **321P2** extending in the first direction DR1 from the first part **321P1**. The second part **321P2** may penetrate the plurality of first material layers **201** and **311** and the plurality of second material layers **313**. The hollowed tunnel insulating layer **325** may be in contact with the first part **321P1** of the hollowed blocking insulating layer **321**. In an embodiment, the hollowed tunnel insulating layer **325** and the hollowed blocking insulating layer **321** may be in contact with each other inside each of the lower layer **301**, the sacrificial layer **305**, and the upper layer **309**. Also, the hollowed tunnel

insulating layer **325** may be locally in contact with the second part **321P2** of the hollowed blocking insulating layer **321**. In an embodiment, the hollowed tunnel insulating layer **325** may be in contact with the second part **321P2** of the hollowed blocking insulating layer **321** at a level at which each of the plurality of material layers **201** and **311** is disposed. Consequently, the data storage pattern **323P** may be excluded between the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325** at a level at which the preliminary doped semiconductor structure **300** is disposed.

(67) After at least one of the channel layer **331**, the core insulating layer **333**, and the capping semiconductor layer **335** is formed inside the hollowed tunnel insulating layer **325**, a slit **353** may be formed to penetrate the preliminary gate stack structure **310**, the gate insulating layer **201**, and the upper layer **309**.

(68) The process of forming the slit **353** may include a first etching process of etching the plurality of first material layers **201** and **311** and the plurality of second material layers **313** and a second etching process of etching the upper layer **309**. The upper layer **309** may serve as an etch stop layer during the first etching process. Accordingly, in an embodiment, the depth of the slit **353** may be precisely controlled.

(69) FIGS. **6A**, **7A**, **8A**, and **9A** are enlarged sectional views illustrating an embodiment of subsequent processes performed after the process shown in FIG. **5D**. FIGS. **6B**, **7B**, **8B**, and **9B** are enlarged sectional views illustrating an embodiment of the subsequent processes performed after the process shown in FIG. **5D**. FIGS. **6A** to **9A** and **6B** to **9B** enlarge and illustrate region A shown in FIG. **5D**.

(70) Referring to FIGS. **6A** and **6B**, a protective structure **350A** or **350B** may be formed along a surface of the slit **353**.

(71) Referring to FIG. **6A**, in an embodiment, the protective structure **350A** may be formed as a double layer including a first protective layer **357** and a second protective layer **359**. The first protective layer **357** may extend along a sidewall and a bottom surface of the slit **353**, and the second protective layer **359** may extend along the first protective layer **357**. The first protective layer **357** may include a material having an etch selectivity with respect to the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325**. In an embodiment, the first protective layer **357** may include nitride. Like the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325**, the second protective layer **359** may include an oxide-based material.

(72) Referring to FIG. **6B**, in an embodiment, the protective structure **350B** may be formed as a single layer. The protective structure **350B** as the single layer may include a material having an etch selectivity with respect to the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325**. In an embodiment, the protective structure **350B** may include nitride. The protective structure **350B** as the single layer may be formed to have a thickness thicker than a thickness of each of the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325**.

(73) Referring to FIGS. **7A** and **7B**, a portion of the protective structure **350A** or **350B** may be removed through a blanket etching process such as etch-back such that the bottom surface of the slit **353** may be exposed. The protective structure **350A** or **350B** may remain on the side wall of the slit **353**.

(74) When the preliminary doped semiconductor structure **300** includes the upper protective layer **307**, the sacrificial layer **305** may be exposed by etching the upper protective layer **307** through the slit **353**.

(75) Referring to FIGS. **8A** and **8B**, the sacrificial layer **305** shown in FIG. **7A** or **7B** may be selectively removed. Accordingly, the lower protective layer **303**, the upper protective layer **307**, and the hollowed blocking insulating layer **321** may be exposed.

(76) Referring to FIGS. **9A** and **9B**, the hollowed blocking insulating layer **321** and the hollowed

tunnel insulating layer **325** may be etched through a region in which the sacrificial layer is removed through an etching process using the protective structure **350A** or **350B** as an etch barrier. While the oxide-based hollowed blocking insulating layer **321** and the oxide-based hollowed tunnel insulating layer **325** are etched, the oxide-based lower protective layer **303** and the oxide-based upper protective layer **307**, which are shown in FIG. **8A** or **8B**, may be removed. Accordingly, the channel layer **331** may be exposed through a horizontal space HSP between the lower layer **301** and the upper layer **309**.

(77) The hollowed blocking insulating layer **321** may be isolated into a first blocking insulating pattern **321A** and a second blocking insulating pattern **321E** by the horizontal space HSP. The hollowed tunnel insulating layer **325** may be isolated into a first tunnel insulating pattern **325A** and a second tunnel insulating pattern **325B** by the horizontal space HSP.

(78) In accordance with an embodiment, a thickness loss of the first protective layer **357** of the protective structure **350A** may be prevented or reduced by the second protective layer **359** shown in FIG. **8A**. Because the second protective layer **359** shown in FIG. **8A** is an oxide-based layer, the second protective layer **359** may be removed while the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325** are etched. Accordingly, the first protective layer **357** may be exposed. In an embodiment, the first protective layer **357** may remain, thereby protecting the preliminary gate stack structure **310**.

(79) In accordance with another embodiment, a partial thickness of the protective structure **350E** having an etch selectivity with respect to the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325** may be lost. However, in an embodiment, the protective structure **350E** is not completely removed but may remain, thereby protecting the preliminary gate stack structure **310**.

(80) FIGS. **10A** and **10B** are sectional views illustrating an embodiment of subsequent processes performed after the process shown in FIG. **9A** or **9B**.

(81) Referring to FIG. **10A**, a doped semiconductor layer may be formed inside the horizontal space HSP shown in FIG. **9A** or **9B**. Accordingly, a doped channel contact layer **361** in contact with the channel layer **331** may be formed. The doped channel contact layer **361** may extend in a direction (e.g., DR2) intersecting the channel layer **331** between the upper layer **309** and the lower layer **301**. The doped channel contact layer **361** may penetrate the sacrificial layer **305** of the preliminary doped semiconductor structure **300** shown in FIG. **5D** and may penetrate a portion of each of the hollowed blocking insulating layer **321** and the hollowed tunnel insulating layer **325**, which are shown in FIG.

(82) Subsequently, the plurality of second material layers **313** of the preliminary gate stack structure **310** shown in FIG. **9A** or **9B** may be exposed by removing the protective structure **350A** or **350B** shown in FIG. **9A** or **9B**. When each of the plurality of second material layers **313** shown in FIG. **9A** or **9B** is a sacrificial material, the plurality of second material layers **313** may be replaced with a plurality of conductive layers **355** through the slit **353**.

(83) Referring to FIG. **10B**, a sidewall insulating layer **360** may be formed on the sidewall of the slit **353** to cover the plurality of conductive layers **355**. Subsequently, a source contact structure **363** may be formed, which is connected to the doped channel contact layer **361**.

(84) FIG. **11** is a block diagram illustrating a configuration of a memory system in accordance with an embodiment of the present disclosure.

(85) Referring to FIG. **11**, the memory system **1100** includes a memory device **1120** and a memory controller **1110**.

(86) The memory device **1120** may be a multi-chip package configured with a plurality of flash memory chips. The memory device **1120** may include a gate stack structure over a doped semiconductor structure, a channel layer which penetrates the gate stack structure and extends to the inside of the doped semiconductor structure, and a blocking insulating pattern and a tunnel insulating pattern, which extend between the channel layer and the doped semiconductor structure

from between the channel layer and the gate stack structure, and have regions in contact with each other. Also, the memory device **1120** may include a data storage pattern which is disposed between the channel layer and the doped semiconductor structure, and is interposed between the blocking insulating pattern and the tunnel insulating pattern at a level at which the gate stack structure is disposed.

(87) The memory controller **1110** controls the memory device **1120**, and may include a Static Random Access Memory (SRAM) **1111**, a Central Processing Unit (CPU) **1112**, a host interface **1113**, an error correction block **1114**, and a memory interface **1115**. The SRAM **1111** is used as an operation memory of the CPU **1112**, the CPU **1112** performs overall control operations for data exchange of the memory controller **1110**, and the host interface **1113** includes a data exchange protocol for a host connected with the memory system **1100**. The error correction block **1114** detects an error included in a data read from the memory device **1120**, and corrects the detected error. The memory interface **1115** interfaces with the memory device **1120**. The memory controller **1110** may further include a Read Only Memory (ROM) for storing code data for interfacing with the host, and the like.

(88) The memory system **1100** configured as described above may be a memory card or a Solid State Disk (SSD), in which the memory device **1120** is combined with the memory controller **1110**. For example, when the memory system **1100** is an SSD, the memory controller **1110** may communicate with the outside (e.g., the host) through one of various interface protocols, such as a Universal Serial Bus (USB) protocol, a Multi-Media Card (MMC) protocol, a Peripheral Component Interconnection (PCI) protocol, a PCI-Express (PCI-E) protocol, an Advanced Technology Attachment (ATA) protocol, a Serial-ATA (SATA) protocol, a Parallel-ATA (PATA) protocol, a Small Computer System Interface (SCSI) protocol, an Enhanced Small Disk Interface (ESDI) protocol, and an Integrated Drive Electronics (IDE) protocol.

(89) FIG. **12** is a block diagram illustrating a configuration of a computing system in accordance with an embodiment of the present disclosure.

(90) Referring to FIG. **12**, the computing system **1200** may include a CPU **1220**, a random access memory (RAM) **1230**, a user interface **1240**, a modem **1250**, and a memory system **1210**, which are electrically connected to a system bus **1260**. When the computing system **1200** is a mobile device, a battery for supplying an operation voltage to the computing system **1200** may be further included, and an application chip set, an image processor, a mobile DRAM, and the like may be further included.

(91) The memory system **1210** may have the same configuration as a memory device **1212** and a memory controller **1211**. The memory device **1212** may be configured identically to the memory device **1120** described above with reference to FIG. **11**. The memory controller **1211** may have the same configuration as the memory controller **1110** described above with reference to FIG. **11**.

(92) In accordance with the present disclosure, in various embodiments, a stack structure of a plurality of material layers may be protected by using a protective structure as a double layer or a single layer, which is disposed on a sidewall of a slit, so that a manufacturing process may be simplified as compared with a case where a protective structure as a multi-layer including three or more layers is introduced.

Claims

1. A method of manufacturing a semiconductor memory device, the method comprising: alternately stacking a plurality of first material layers and a plurality of second material layers in a first direction over a preliminary doped semiconductor structure; forming a hollowed blocking insulating layer penetrating the plurality of first material layers and the plurality of second material layers, the hollowed blocking insulating layer extending to the inside of the preliminary doped semiconductor structure; forming a plurality of data storage patterns on levels where the plurality

of second material layers are disposed, wherein an inner wall of the hollowed blocking insulating layer is partially covered by the plurality of data storage patterns; forming a hollowed tunnel insulating layer to cover the plurality of data storage patterns and the inner wall of the hollowed blocking insulating layer; forming a channel layer inside the hollowed tunnel insulating layer; forming a slit penetrating the plurality of first material layers and the plurality of second material layers; forming a protective structure as one of a double layer and a single layer on a sidewall of the slit; and forming a doped channel contact layer penetrating a portion of the preliminary doped semiconductor structure in a direction intersecting the channel layer, the doped channel contact layer being in contact with the channel layer.

2. The method of claim 1, wherein the double layer includes a first protective layer extending along the side all of the slit and a second protective layer extending along the first protective layer, wherein the first protective layer includes a material having an etch selectivity with respect to the hollowed blocking insulating layer and the hollowed tunnel insulating layer, and wherein each of the hollowed blocking insulating layer, the hollowed tunnel insulating layer, and the second protective layer includes oxide.

3. The method of claim 2, wherein the first protective layer includes nitride.

4. The method of claim 1, wherein the single layer includes a material having an etch selectivity with respect to the hollowed blocking insulating layer and the hollowed tunnel insulating layer.

5. The method of claim 4, wherein the single layer includes nitride.

6. The method of claim 1, wherein the preliminary doped semiconductor structure includes a lower layer, a sacrificial layer, and an upper layer; which are stacked in the first direction, and wherein the hollowed blocking insulating layer and the hollowed tunnel insulating layer are in contact with each other inside each of the lower layer, the sacrificial layer, and the upper layer.

7. The method of claim 6, wherein the slit penetrates the upper layer, and wherein the forming of the doped channel contact layer includes: removing the sacrificial layer through the slit such that the hollowed blocking insulating layer is exposed; etching the hollowed blocking insulating layer and the hollowed tunnel insulating layer through a region in which the sacrificial layer is removed through an etching process using the protective structure as an etch barrier; and forming a doped semiconductor layer in a region in which the hollowed blocking insulating layer and the hollowed tunnel insulating layer are etched and a region in which the sacrificial layer is removed.

8. The method of claim 1, wherein the hollowed blocking insulating layer includes a first part inside the preliminary doped semiconductor structure and a second part extending to penetrate the plurality of first material layers and the plurality of second material layers from the first part, and wherein the hollowed tunnel insulating layer is in contact with the first part of the hollowed blocking insulating layer.

9. The method of claim 8, wherein the hollowed tunnel insulating layer is in contact with the second part of the hollowed blocking insulating layer at a level at which each of the plurality of first material layers is disposed.

10. The method of claim 1, further comprising: replacing the plurality of second material layers with a plurality of conductive layers through the slit; forming a sidewall insulating layer on the sidewall of the slit to cover the plurality of conductive layers; and forming a source contact structure connected to the doped channel contact layer inside the slit.

11. A semiconductor memory device comprising: a doped semiconductor structure; a gate stack structure including a plurality of conductive layers stacked in a first direction over the doped semiconductor structure; a channel layer including a first part and a second part extending in the first direction from the first part to penetrate the gate stack structure; a first tunnel insulating pattern extending along an outer wall of the second part of the channel layer; a plurality of data storage patterns interposed between the plurality of conductive layers and the first tunnel insulating pattern; and a first blocking insulating pattern interposed between each of the plurality of data storage pattern and the gate stack structure, wherein at least a portion of the first part is located inside the

doped semiconductor structure, wherein the doped semiconductor structure includes a contact surface in contact with the channel layer, and wherein the first blocking insulating pattern extends to a first region between a lowermost conductive layer adjacent to the doped semiconductor structure among the plurality of conductive layer and the contact surface, and is in contact with the first tunnel insulating pattern in the first region.

12. The semiconductor memory device of claim 11, wherein the first blocking insulating pattern extends to a second region between the plurality of conductive layers adjacent to each other in the first direction, and is in contact with the first tunnel insulating pattern in the second region.

13. The semiconductor memory device of claim 11, wherein the doped semiconductor structure includes: a doped channel contact layer having a sidewall corresponding to the contact surface; an upper layer between the doped channel contact layer and the gate stack structure; and a lower layer spaced apart from the upper layer with the doped channel contact layer interposed therebetween, and wherein the first tunnel insulating pattern and the first blocking insulating pattern are in contact with each other between the upper layer and the channel layer.

14. The semiconductor memory device of claim 13, further comprising: a second tunnel insulating pattern between the first part of the channel layer and the lower layer; and a second blocking insulating pattern interposed between the second tunnel insulating pattern and the lower layer, the second blocking insulating pattern being in contact with the second tunnel insulating pattern.

15. The semiconductor memory device of claim 11, wherein the gate stack structure further includes a plurality of interlayer insulating layers alternately disposed in the first direction with the plurality of conductive layers, and wherein the first blocking insulating pattern and the first tunnel insulating pattern are in contact with each other between the plurality of interlayer insulating layers and the second part of the channel layer.
